Yashas K

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Objective

2nd-year engineering student with independent research experience in data structure design and information retrieval. Seeking a software engineering internship where I can contribute to retrieval, indexing, or content pipeline infrastructure while continuing to develop my implementation skills. Honest about my current skill level: I design systems and drive research direction; I am actively learning Rust and Python to fully own my implementations.

Skills

Strong	Systems design, algorithm analysis, measurement methodology, technical writing, research documentation
Learning	Rust (beginner, actively studying via The Rust Programming Language + Rustlings), Python (intermediate)
Tools	Git, GitHub, LaTeX, basic shell scripting
Concepts	Binary tries, LZ77 compression, LOUDS succinct encoding, suffix trees (theoretical), FAISS/ANN (theoretical), RAG architectures

Featured Project: AXIOM — Binary Substring Graph for Zero-Error Retrieval

github.com/Yashas-K-Gangatkar/axiom-3d · October 2026

Independent research project on exact-match retrieval. Designed the architecture; implementation was developed with AI assistance.

My role (what I can defend in an interview):

- Designed the three-layer architecture: V1 prefix trie, V2 LZ77 macro graph, V3.1 LOUDS succinct trie
- Defined the honest measurement methodology: real Gutenberg text at 1/10/100 MB, 5 adversarial fake queries per size
- Specified the trade-offs: 212× bloat in V1, 94% reduction in V3.1, $O(N \cdot L)$ substring search complexity
- Wrote the IEEE-format paper draft (8 pages, 14 references), including the honest limitations section
- Drove the research direction: zero-error verification as a complement to ANN-based retrieval in RAG pipelines

What I am currently learning to fully own:

- Rust ownership, borrowing, and lifetimes (to read and modify the implementation)
- LOUDS rank/select implementation details (to extend V3.1)
- Ukkonen's suffix tree construction (the next planned upgrade)

Honest limitations of the project:

- V2 is 38× worse than `xz -9` on compression ratio (no entropy coding)
- Substring search is $O(N \cdot L)$, not $O(L)$ — suffix-tree upgrade is future work
- Single-threaded build (no parallelism yet)

Education

Bachelor of Engineering, Computer Science — 2024 – 2028 (expected)
Engineering College, Bangalore, India — 2nd year (current)

- Independent research focus: data structures, information retrieval, ML infrastructure

- Currently studying: Rust (via The Rust Programming Language book), algorithms (CLRS), ML systems design

Currently Learning (October 2026)

- **Rust**: Working through The Rust Programming Language (chapters 1–10 complete) and Rustlings exercises. Goal: rewrite AXIOM V1 from scratch in Rust by December 2026.
- **Algorithms**: Reviewing CLRS chapters on tries, suffix trees, and LZ77 to deepen theoretical understanding of AXIOM's design.
- **ML systems**: Reading FAISS paper and HNSW paper to understand the ANN landscape AXIOM is positioned against.

What I'm Looking For

- **Summer 2027 internship** (3 months, May–August) in software engineering, with focus on:
  - Backend systems / infrastructure
  - Data engineering / pipelines
  - Search / retrieval / indexing
- **Open to**: ETH Student Summer Research Fellowship (apply November 2026), Disney Streaming Zurich SWE intern, Microsoft Zurich SWE intern, or any role where I can both contribute and continue learning
- **Not yet ready for**: Senior SWE roles, roles requiring 2+ years of Rust production experience, roles requiring ML research scientist credentials

How I Work

- I design systems and document them rigorously, including their failure modes
- I use AI assistance (Z.ai/GLM, GitHub Copilot) for implementation, then study the generated code to learn from it
- I am honest about what I know and what I'm learning — I will not claim a skill I cannot defend
- I write things down: every project I work on has a paper draft, a README, and a reproducibility guide

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All claims in this resume are verifiable in my GitHub repositories. I am happy to walk through any project in an interview, including the parts I am still learning.